CCTV Engineering Report

Technical Engineering Report

DRAFT - For engineering review; site verification required

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1. Project Overview, Scope, and Design Basis

This report establishes the design basis and implementation framework for a closed-circuit television (CCTV) system serving the project site. The system is intended to provide security monitoring, incident detection, event verification, and evidentiary video recording across identified operational, access, perimeter, and critical-asset areas. The final site name, address, facility type, operating hours, threat profile, and required retention period are to be confirmed and recorded as TBD until approved project information is available.

The project scope comprises CCTV design development, camera coverage planning, equipment selection, network and power coordination, recording and storage provisions, installation requirements, system configuration, cybersecurity controls, testing, commissioning, and handover documentation. The scope includes fixed cameras and any additional device types required by the approved risk assessment, such as panoramic, varifocal, infrared, or specialist cameras. Monitoring workstations, video management software, network switches, recording servers or appliances, displays, uninterruptible power supplies, enclosures, cabling, and associated licenses are included where required by the final system architecture. Quantities, device models, cable distances, storage capacities, and equipment locations remain TBD unless verified by approved drawings or site surveys.

The design shall be based on the security objectives established by the project owner and the physical conditions identified through document review and site verification. Camera selection and positioning shall support the intended viewing objective for each location, such as detection, recognition, identification, situational awareness, or overview monitoring. Field of view, lighting conditions, backlighting, glare, privacy considerations, mounting height, environmental exposure, maintenance access, and potential obstructions shall be assessed before final installation. Camera coverage shall not be considered complete solely on the basis of nominal resolution; it shall be validated against the required operational outcome.

The CCTV system shall be designed as an integrated IP-based solution unless the approved project requirements state otherwise. Equipment shall be compatible with the selected video management platform and shall support required network, power, recording, time-synchronization, access-control, and alarm interfaces. Power-over-Ethernet (PoE) availability, switch capacity, network segmentation, bandwidth, and resilience shall be confirmed during detailed design. Electrical supplies and standby-power requirements shall be coordinated with the electrical and information technology disciplines.

Applicable legislation, authority requirements, information-security policies, privacy obligations, and recognized industry standards shall be identified and confirmed by the responsible project stakeholders before procurement. Any assumptions, exclusions, unresolved interfaces, or deviations from the approved design basis shall be documented in the project risk and action registers. Final coverage, equipment quantities, performance requirements, and acceptance criteria shall remain subject to owner approval and successful commissioning.

2. Site Security Objectives and Coverage Requirements

The CCTV system shall support the site’s security objectives by providing effective deterrence, timely detection, reliable assessment of security events, and evidentially useful recording. The design shall enable authorized personnel to identify relevant activity, review incidents, verify alarms, and support post-event investigations. CCTV shall complement, rather than replace, physical security measures such as perimeter barriers, access control, lighting, intrusion detection, security patrols, and site operating procedures.

Coverage shall be developed from a current site plan, security risk assessment, operational brief, and agreed camera positioning review. The final design shall identify all areas requiring surveillance, including site approaches, vehicle and pedestrian entrances, perimeter sections, building entrances and exits, loading and service areas, parking areas, circulation routes, critical plant or infrastructure, restricted zones, and locations containing high-value assets. Areas that cannot be effectively monitored shall be recorded as design exclusions or residual risks, with justification and compensating measures identified where applicable.

Each camera shall have a defined viewing objective. Depending on the risk and operational need, this may be general monitoring, activity detection, facial recognition, vehicle identification, license-plate capture, access-point verification, or protection of a specific asset. Camera selection, mounting position, field of view, lens type, resolution, frame rate, illumination capability, and image retention shall be appropriate to the stated objective. A camera shall not be considered suitable solely because the target area appears within its nominal field of view; the required image detail, lighting conditions, scene geometry, and expected target movement shall also be considered.

Coverage shall account for day and night operation, glare, shadows, weather exposure, foliage growth, obstructions, changing vehicle positions, and foreseeable changes to the site layout. Critical views shall be designed to minimize blind spots and avoid reliance on a single camera where loss of that view would materially affect security operations. Camera placement shall also consider privacy, proportionality, neighboring properties, public areas, worker welfare, safe maintenance access, and protection against tampering or accidental impact.

The system shall provide consistent time reference, controllable access to live and recorded video, event search facilities, and recording retention in accordance with the approved operational, legal, and organizational requirements. Retention duration, monitoring arrangements, alarm integration, privacy masking, user roles, and export procedures shall be confirmed by the project authority and recorded as TBD where not yet defined. Final coverage acceptance shall be based on documented verification of each camera’s intended viewing objective, image quality, field of view, lighting performance, recording, and operational availability.

3. CCTV System Architecture and Equipment Schedule

The proposed CCTV system shall use a network-based architecture comprising IP cameras, Power over Ethernet (PoE) network switches, a video management system (VMS), recording and storage infrastructure, operator workstations, and associated network, power, and mounting accessories. Cameras shall transmit video over the site data network to the designated recording platform. The final topology, equipment locations, addressing scheme, and network segmentation shall be confirmed during detailed design and recorded as TBD where project information is not yet available.

The architecture shall support centralised monitoring, recording, playback, export, user administration, event search, and system health monitoring. Camera streams shall be configured in accordance with the approved viewing objectives and available network and storage capacity. Where supported by the selected equipment, the system should provide motion detection, tamper alarms, video-loss notification, privacy masking, time synchronisation, and configurable retention policies. Any analytics functions shall be enabled only where their performance requirements and operational use have been defined and verified.

Cameras shall be selected according to location, lighting conditions, field of view, identification requirements, environmental exposure, and mounting arrangement. Equipment installed outdoors or in exposed areas shall have an enclosure and environmental rating suitable for the location. Internal cameras shall be selected to suit the architectural environment while maintaining the required image quality. Lens type, resolution, infrared capability, wide dynamic range, low-light performance, and image settings shall be confirmed against the approved camera schedule. Exact camera models and quantities remain TBD unless established by the project design.

PoE switches shall provide sufficient powered ports, uplink capacity, and operational resilience for the connected cameras, with allowance for identified expansion requirements where applicable. Switches shall be installed in suitable communications locations and connected to the site network using approved cabling and termination practices. The design shall identify any requirement for fibre uplinks, intermediate distribution points, media converters, surge protection, or local power supplies. These requirements, together with cabinet locations and port allocations, are TBD pending network and site coordination.

The recording platform shall comprise either a dedicated network video recorder, server-based VMS, or approved equivalent. It shall support the required camera count, recording modes, resolution, frame rate, retention period, user access controls, and evidence export functions. Final storage capacity and recording configuration shall be calculated from confirmed camera parameters and retention requirements; no capacity or retention value shall be assumed at this stage.

The equipment schedule shall identify each equipment category, quantity, manufacturer, model, location, power requirement, network interface, and verification status. Unconfirmed items shall be marked TBD. The schedule shall be updated during detailed design, installation, commissioning, and handover to reflect approved equipment and any authorised changes.

4. Camera Schedule Template: Camera ID, Location, Viewing Objective, Lens/Resolution, PoE, and Verification Status

The camera schedule shall be used as the controlled register for identifying each CCTV device, its intended coverage, technical requirements, and verification status. Each camera should be assigned a unique Camera ID that remains consistent across drawings, network records, recording-system configuration, labels, test sheets, and handover documentation. Locations shall identify the relevant building, level, room, elevation, entrance, service area, or external zone sufficiently to support installation and maintenance.

The viewing objective shall describe the operational purpose of the camera, such as identification at an access point, recognition of activity in a restricted area, observation of circulation routes, monitoring of vehicle movement, or general situational awareness. Lens selection and resolution shall be confirmed against the required field of view, target distance, lighting conditions, image detail, and camera mounting arrangement. Where these design inputs have not yet been approved, the schedule shall retain a TBD entry rather than implying a completed selection.

PoE requirements shall be coordinated with the network and power design. The schedule should identify the intended PoE provision or class where known, together with any requirement for local power, an extended-reach solution, an outdoor-rated connection, or a separate power budget review. Verification status shall distinguish design information from field confirmation. Suitable entries may include “Not Started,” “Design Review,” “Installed—Pending Test,” “Verified,” or “Exception—Action Required.” A camera shall not be recorded as fully verified until its physical installation, view, focus, connectivity, recording, time synchronization, and applicable image-quality requirements have been checked and documented.

4. Camera Schedule Template: Camera ID, Location, Viewing Objective, Lens/Resolution, PoE, and Verification Status — Table

Camera ID	Location	Viewing Objective	Lens/Resolution	PoE	Verification Status
CAM-TBD-001	TBD	TBD; confirm target area and operational purpose	TBD	TBD	Not Started
CAM-TBD-002	TBD	TBD; confirm target area and operational purpose	TBD	TBD	Not Started
CAM-TBD-003	TBD	TBD; confirm target area and operational purpose	TBD	TBD	Not Started
CAM-TBD-004	TBD	TBD; confirm target area and operational purpose	TBD	TBD	Not Started
CAM-TBD-005	TBD	TBD; confirm target area and operational purpose	TBD	TBD	Not Started
CAM-TBD-006	TBD	TBD; confirm target area and operational purpose	TBD	TBD	Not Started
CAM-TBD-007	TBD	TBD; confirm target area and operational purpose	TBD	TBD	Not Started
CAM-TBD-008	TBD	TBD; confirm target area and operational purpose	TBD	TBD	Not Started

The completed schedule shall be reviewed against the approved security objectives, camera layout drawings, equipment schedule, network switch-port allocation, and recording configuration. Any change to a camera's location, field of view, lens, resolution, PoE arrangement, or status shall be traceable through the project change-control process. Columns may be expanded to include mounting height, orientation, IP address, switch port, enclosure type, commissioning reference, and outstanding actions where required by the project.

5. Network, Power, Recording, and Storage Design

The CCTV system shall use a dedicated, logically separated network wherever practical. Cameras should connect to managed PoE access switches, with uplinks to the video management system (VMS), recording servers, or network video recorders (NVRs) through a secured distribution layer. The CCTV network shall be isolated from general business traffic using dedicated VLANs and controlled routing. Inter-VLAN communication shall be limited to required services, such as camera management, time synchronization, authentication, monitoring, and VMS or NVR communication. Network topology, switch locations, uplink capacity, and redundancy requirements shall be confirmed during detailed design; project-specific values remain TBD.

PoE switches shall be selected to support the connected camera types, auxiliary infrared illuminators, microphones, heaters, or other powered accessories identified in the equipment schedule. Each switch shall have sufficient PoE port capacity and power budget for its connected devices, including an allowance for startup and peak demand. PoE standards shall be compatible with the equipment manufacturer's requirements. Switches should provide port-level monitoring, fault reporting, loop protection, and configuration backup. Unused ports shall be disabled or otherwise restricted, and camera-facing ports shall be configured only for the services required by the CCTV system.

The CCTV network and recording equipment shall be supplied from conditioned power supported by an uninterruptible power supply (UPS), where required by the project continuity objectives. UPS capacity, autonomy, distribution-board source, and generator interface are TBD unless established by the electrical design. Critical switches, recording servers, storage systems, and network equipment shall be connected in accordance with the approved power schedule. Surge protection, earthing, environmental conditions, and equipment-rack ventilation shall comply with applicable electrical and manufacturer requirements. Power circuits shall be clearly labelled and coordinated with facilities operations.

Recording shall be performed by the approved VMS or NVR platform using the final camera schedule, recording profiles, frame rates, resolutions, compression settings, and event rules. Continuous, scheduled, motion-based, or alarm-triggered recording shall be assigned by camera according to the security objectives and operational requirements. Retention duration, recording availability target, archive requirements, and evidential export format are TBD. Storage calculations shall account for the number of cameras, encoded bitrate, recording hours per day, retention period, metadata, system overhead, RAID or other resilience method, and required free capacity. The design shall document usable capacity separately from raw disk capacity.

System clocks shall be synchronized through an approved time source to maintain consistent timestamps across cameras, servers, and exported evidence. Recording health, storage utilization, camera connectivity, PoE status, and UPS alarms shall be monitored and reported to authorized personnel. Any redundancy, failover, backup, or off-site replication requirements shall be confirmed and documented before procurement.

6. Installation, Configuration, and Cybersecurity Requirements

Installation shall be completed in accordance with the approved drawings, equipment manufacturer instructions, applicable electrical and communications codes, and site safety procedures. Camera locations, mounting heights, fields of view, cable routes, network outlets, and equipment cabinet positions shall be verified against the approved design before permanent installation. Any obstruction, access limitation, environmental concern, or deviation from the design shall be recorded for review rather than resolved through undocumented field changes.

Cameras shall be mounted securely on suitable structural supports, with vibration, glare, weather exposure, and tampering considered. Outdoor devices and associated enclosures shall have environmental ratings appropriate to their locations. Cable entries shall be sealed, and exposed cabling shall be protected against mechanical damage and unauthorized access. Category cabling shall be installed, terminated, labeled, and tested in accordance with the project cabling requirements. Each cable and device shall have a unique identifier matching the camera schedule and network documentation.

Initial configuration shall use approved device names, static addressing or reserved addressing as defined by the network design, correct time zone, and synchronized time from the authorized time source. Default usernames, passwords, certificates, and unnecessary services shall be changed or disabled before connection to the operational network. Administrative access shall be limited to authorized personnel, use individual accounts where supported, and apply least-privilege permissions. Passwords shall meet the project’s approved complexity and rotation requirements, which are TBD where not yet established.

The video management system, recording servers, storage systems, switches, and cameras shall be configured to the approved recording, retention, resolution, frame-rate, event, and export requirements. Values not defined in the approved design shall remain TBD and shall not be assumed during installation. Camera views shall be adjusted to achieve the stated viewing objective while avoiding unnecessary capture of adjacent private or restricted areas. Privacy masking shall be applied where required and documented.

Cybersecurity controls shall include network segmentation or equivalent access control, restricted management interfaces, secure remote access, current supported firmware, and controlled update procedures. Devices shall not be exposed directly to the public internet. Firewall rules shall permit only required services between cameras, recording platforms, workstations, storage, authentication, and time services. Antivirus or endpoint protection shall be applied to applicable servers and workstations, subject to system compatibility. Configuration backups shall be protected from unauthorized modification and stored in an approved location.

Before acceptance, installers shall verify device identification, image quality, focus, coverage, infrared or low-light operation where applicable, recording continuity, time synchronization, alarm and event functions, user access, failover behavior, and recovery after power or network interruption. Test results shall be entered in the commissioning checklist, with unresolved deficiencies assigned an owner and status. Configuration changes made after testing shall be documented and retested as necessary.

7. Commissioning Checklist Template: Item, Test Method, Expected Result, Actual Result, and Status

Commissioning shall verify that the installed CCTV system conforms to the approved design, equipment schedule, configuration requirements, and applicable project specifications. Each checklist item shall be tested using the stated method, with the result recorded against the relevant camera, device, network segment, recording server, or monitoring workstation. Measurements, test evidence, and pass/fail determinations shall be entered only after the activity has been completed. Where information is unavailable or testing remains outstanding, record TBD rather than assuming compliance.

The commissioning record should include the tester’s name, date, equipment or camera reference, and supporting evidence where applicable, such as screenshots, exported configuration files, test photographs, cable certification, or alarm-event logs. Any failed or partially completed item shall be assigned an action owner and target completion date in the project action register. Retesting shall be recorded as a new result or clearly identified revision to the original entry.

7. Commissioning Checklist Template: Item, Test Method, Expected Result, Actual Result, and Status — Table

Item	Test Method	Expected Result	Actual Result	Status
Camera identity and location	Compare physical label, camera ID, and installed location with the approved camera schedule and drawings.	Camera is correctly identified, labelled, and installed at the approved location.	TBD	TBD
Camera image and field of view	View live image from the approved client or VMS and compare with the viewing objective and design coverage.	Image is available, correctly oriented, adequately focused, and provides the required coverage without unacceptable obstruction.	TBD	TBD
Image quality and settings	Review live and recorded images under representative lighting conditions; verify resolution, frame rate, compression, and time settings against approved values.	Image quality and configuration meet the approved design criteria.	TBD	TBD
Camera focus, housing, and environmental condition	Conduct visual inspection and, where applicable, verify heater, fan, weatherproofing, and physical security.	Camera and housing are secure, undamaged, sealed, and suitable for the installed environment.	TBD	TBD
PoE and network connectivity	Verify switch port assignment, link status, power delivery, addressing, and connectivity to the designated network services.	Camera receives stable power and communicates only through the approved network path.	TBD	TBD
Video recording	Generate a controlled test event and search for the corresponding recording using the VMS.	Video is recorded, time-stamped, retrievable, and associated with the correct camera ID.	TBD	TBD
Retention and storage	Review configured retention, storage allocation, and system health indicators against the approved design.	Storage capacity, retention settings, and system health satisfy project requirements.	TBD	TBD
Time synchronisation	Compare camera, VMS, server, and monitoring workstation time against the approved time source.	All relevant devices maintain a consistent, synchronised time reference.	TBD	TBD
Alarm and event functions	Trigger applicable analytics, tamper, motion, input, or system alarms under controlled conditions.	Required events are generated, displayed, logged, and escalated to designated users.	TBD	TBD
User access and cybersecurity controls	Test authorised and unauthorised login scenarios and review roles, passwords, services, and audit logging.	Access is role-based, default credentials are removed or changed, and security events are logged.	TBD	TBD
Backup, recovery, and fault reporting	Review backup configuration and simulate an approved component or communication fault where safe to do so.	Faults are reported, recovery procedures operate as designed, and no critical data is lost.	TBD	TBD
Documentation and outstanding defects	Review drawings, schedules, configurations, test records, and the defect register.	Handover records are complete, traceable, and all outstanding items are identified.	TBD	TBD

8. Handover Documentation, Outstanding Items, and TBD Project Values

Handover shall be completed only after installation, configuration, commissioning, and documentation activities have been reviewed against the approved design basis and project requirements. The handover package shall include the final camera schedule, marked-up or as-built drawings, equipment datasheets, network and addressing information, recording and retention settings, configuration backups, licence information, and relevant manufacturer documentation. Any changes made during installation shall be incorporated into the final records, with revisions identified and approved through the project change-control process.

The handover package shall also include commissioning records for each camera and associated system function. These records shall identify the test method, expected result, actual result, status, date, and responsible tester. Verification should cover camera identification, field of view, image quality, focus, time and date synchronisation, infrared or illumination performance where applicable, video recording, playback, export, alarms, user access, network connectivity, storage operation, and loss-of-power or communications recovery. No test shall be recorded as complete unless supported by an observed result or approved evidence.

Outstanding items shall be maintained in a controlled action register until closure. Each item should include a unique reference, description, responsible party, required action, priority, target completion date, and closure evidence. Typical outstanding items may include final camera positioning approval, resolution of access or containment constraints, completion of labels, delivery of configuration backups, cybersecurity review, operator training, integration with other systems, rectification of defects, and approval of as-built documentation. Items not affecting safe operation may be accepted temporarily only where the project owner formally records the limitation, interim control, and agreed completion date.

Project values not established by the available design or site information shall remain identified as TBD and shall not be inferred from typical practice. Values requiring confirmation may include camera quantities, final camera locations, lens selections, image-resolution requirements, PoE classes, switch capacities, IP addressing, VLAN details, bandwidth allowances, recording frame rates, retention periods, storage capacity, UPS autonomy, time-source configuration, user roles, alarm inputs and outputs, and system integration points. Each TBD value shall have an identified owner and a required confirmation stage.

Before final acceptance, the project owner shall confirm that all required documentation is complete, all critical defects are closed, operational staff have received appropriate instruction, and remaining TBD values have either been resolved or formally accepted with documented assumptions and constraints. Handover acceptance shall be recorded by the authorised project representatives, with the final document revision and outstanding-item status retained as part of the project record.